AI – RESUME ANALYSER

Prepared in the partial fulfillment of the Summer Internship Program on AWS

AT

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Thank you.

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ABSTRACT

This documentation presents the design and implementation of the AI Resume Analyzer, a fully serverless, AI-powered platform built on Amazon Web Services (AWS) that automates resume screening and career-readiness feedback. The system allows a candidate to upload a resume (PDF or DOCX), paste a target job description, and receive an instant, structured evaluation covering an overall resume quality score, a job-fit match percentage, matched and missing skills, strengths, weaknesses, actionable improvement suggestions, and recommended courses to close identified skill gaps.

The platform is built entirely on managed, pay-per-use AWS services. Resumes are uploaded directly to Amazon S3 via presigned URLs, text is extracted using Amazon Textract (with DOCX files converted to PDF in-Lambda via a bundled LibreOffice layer), and the extracted content is structured into contact, education, experience, skills, and certification fields using a Lambda-based parser backed by a curated skills taxonomy. The structured resume and the job description are then passed to Amazon Bedrock, where an Anthropic Claude foundation model produces a strict, schema-validated JSON evaluation. Results are persisted in Amazon DynamoDB, exposed through a Cognito-authenticated Amazon API Gateway REST API, and rendered in a React single-page application.

By combining Amazon Textract, Amazon Bedrock, AWS Lambda, DynamoDB, Cognito, and API Gateway, the AI Resume Analyzer eliminates the time-consuming and inconsistent nature of manual resume screening while keeping candidate data entirely within the AWS account boundary. The architecture scales automatically with demand, incurs no idle cost, and lays a practical foundation for future enhancements such as recruiter-facing dashboards, bulk processing, and multi-language support.

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1. INTRODUCTION

In today's highly competitive job market, organizations receive hundreds or even thousands of resumes for a single job opening. Human Resource (HR) professionals spend a significant amount of time manually reviewing these resumes to identify suitable candidates. This traditional process is time-consuming, labor-intensive, and often leads to inconsistent evaluations due to human bias and fatigue. As companies continue to receive an increasing number of applications, the need for intelligent and automated resume screening solutions has become more important than ever.

Artificial Intelligence (AI) and cloud computing technologies have transformed the recruitment industry by enabling organizations to automate repetitive tasks and improve hiring efficiency. Modern Applicant Tracking Systems (ATS) help recruiters filter resumes based on predefined criteria, but many traditional systems rely heavily on keyword matching and often fail to evaluate the overall quality of a candidate's profile. Recent advancements in Large Language Models (LLMs) have introduced new opportunities to analyze resumes more intelligently by understanding context, evaluating skills, identifying experience, and providing meaningful recommendations.

Cloud platforms such as Amazon Web Services (AWS) provide highly scalable, secure, and cost-effective services that allow developers to build intelligent applications without managing physical servers. By combining AI capabilities with serverless cloud architecture, organizations can develop applications that automatically process resumes, extract meaningful information, compare candidate profiles with job descriptions, and generate detailed analytical reports within seconds.

The AI-Powered Resume Analyzer presented in this project leverages AWS serverless services and Generative AI to automate the complete resume evaluation process. The system enables users to upload resumes in PDF or DOCX format, enter a target job description, and receive an instant analysis report containing ATS score, job match percentage, identified skills, missing skills, strengths, weaknesses, improvement suggestions, and recommended learning resources. This intelligent approach significantly reduces manual effort while providing candidates with valuable career guidance and improving the overall recruitment process.

2. METHODOLOGY

The development of the **AI-Powered Resume Analyzer** followed a structured Software Development Life Cycle (SDLC) combined with cloud-native development practices to ensure a scalable, secure, and reliable application. The methodology consisted of several phases including requirements gathering, system design, implementation, testing, deployment, and continuous refinement. Each phase focused on achieving the project objectives while leveraging Amazon Web Services (AWS) serverless technologies and Generative AI for intelligent resume analysis.

The following subsections explain each phase of the development process.

2.1 Requirements Gathering

The first phase of the project involved identifying the functional and non-functional requirements of an intelligent resume analysis platform. The objective was to understand the challenges faced by both recruiters and job seekers during the resume screening process and to determine how Artificial Intelligence and cloud computing could simplify these tasks.

Functional requirements included secure user authentication, resume upload functionality, automatic text extraction, resume parsing, AI-based evaluation, job description comparison, ATS score calculation, report generation, and resume history management. The system was designed to support PDF and DOCX resume formats while maintaining compatibility with multiple job roles.

Non-functional requirements included scalability, security, reliability, high availability, low operational cost, fast response time, and modular architecture. Since the application is deployed on AWS, the infrastructure was designed to automatically scale based on workload without requiring manual server management.

After analyzing these requirements, a serverless architecture using AWS services was selected as the most suitable solution due to its flexibility, security, and cost-effectiveness.

2.2 System Design

Based on the identified requirements, a cloud-native architecture was designed using Amazon Web Services (AWS). The architecture follows a serverless model where each component performs a dedicated task while communicating through secure APIs.

The frontend application was developed using React.js to provide an interactive user interface for registration, login, resume upload, and viewing analysis reports. User authentication is managed through Amazon Cognito, which provides secure signup, email verification, login, password recovery, and JWT-based authorization.

The backend consists of multiple AWS Lambda functions responsible for generating upload URLs, extracting resume text, parsing resume content, invoking Amazon Bedrock for AI analysis, generating structured reports, and retrieving analysis history. Amazon API Gateway exposes these backend services as secure REST APIs.

Amazon S3 stores uploaded resumes, Amazon Textract extracts textual information from documents, Amazon DynamoDB stores user data and analysis reports, and Amazon Bedrock performs intelligent resume evaluation using the Anthropic Claude foundation model. Amazon CloudWatch monitors application logs and system performance.

This modular design improves maintainability, scalability, and fault tolerance while minimizing infrastructure management.

2.3 Implementation

The implementation phase transformed the architectural design into a fully functional AI-powered web application. The development process was divided into frontend development, backend service implementation, cloud service integration, and AI model integration.

The frontend was developed using React.js and JavaScript to provide an intuitive user experience. Authentication pages, resume upload interface, report visualization, and history management modules were implemented.

On the backend, multiple AWS Lambda functions were developed using Python. Each Lambda function performs a specific responsibility, enabling better modularity and easier maintenance. The Upload URL Lambda generates secure pre-signed URLs for Amazon S3 uploads. The Text Extraction Lambda invokes Amazon Textract to extract resume content. The Resume Parser Lambda structures the extracted text into predefined sections such as education, work experience, technical skills, certifications, and projects.

The AI Analyzer Lambda sends the structured resume and target job description to Amazon Bedrock, where the Anthropic Claude model evaluates the candidate's profile. The Report Generator Lambda formats the AI response into a user-friendly report containing ATS score, job match percentage, strengths, weaknesses, missing skills, improvement suggestions, and recommended learning resources. The Get History Lambda retrieves previous analysis reports from Amazon DynamoDB for authenticated users.

Each component was integrated using Amazon API Gateway and secured through JWT authentication provided by Amazon Cognito.

2.4 Database Management

A NoSQL database architecture was selected using Amazon DynamoDB to efficiently store user information, resume metadata, parsed resume content, and AI-generated analysis reports.

Each analysis report contains details such as Resume ID, User ID, Upload Timestamp, ATS Score, Job Match Percentage, Extracted Skills, Missing Skills, Strengths, Weaknesses, Improvement Suggestions, and AI-generated recommendations. DynamoDB enables fast read and write operations while automatically scaling according to application demand.

Data consistency is maintained through validation within the Lambda functions before storing records. Since DynamoDB is fully managed by AWS, no manual database administration or server provisioning is required, reducing operational complexity while improving application reliability.

2.5 Testing

Comprehensive testing was performed throughout the development process to ensure the correctness, reliability, and performance of the application.

Unit testing was conducted for individual Lambda functions to verify resume upload, text extraction, parsing, AI analysis, report generation, and history retrieval. Integration testing validated the interaction between AWS services including Amazon S3, Textract, Lambda, API Gateway, Bedrock, DynamoDB, and Cognito.

Functional testing confirmed that users could successfully register, verify email addresses, log in securely, upload resumes, analyze job compatibility, and retrieve historical reports. Error handling mechanisms were tested using invalid file formats, corrupted documents, unauthorized API requests, and incomplete job descriptions.

Performance testing evaluated system response time for resumes of different sizes and verified that serverless architecture automatically handled concurrent requests without performance degradation.

2.6 Deployment

The application was deployed entirely on Amazon Web Services using AWS Serverless Application Model (AWS SAM). Infrastructure components including Lambda functions, API Gateway, IAM roles, DynamoDB tables, and S3 buckets were defined as Infrastructure as Code templates, enabling repeatable and automated deployment.

AWS CLI and AWS SAM CLI were used to build, package, and deploy the application. IAM roles were configured following the principle of least privilege, ensuring that each AWS service had only the permissions necessary to perform its designated tasks.

Amazon CloudWatch was configured to collect logs and monitor Lambda execution, allowing developers to quickly identify errors and optimize application performance. The serverless deployment model enables automatic scaling, high availability, and reduced operational cost because resources are consumed only when requests are processed.

2.7 Iterative Refinement

Throughout the project development lifecycle, continuous improvements were made based on testing results, mentor feedback, and performance analysis.

The AI prompts used for Amazon Bedrock were refined to improve the quality and consistency of resume evaluation. Resume parsing algorithms were enhanced to recognize a wider variety of resume formats and improve skill extraction accuracy. Error handling mechanisms were strengthened to support invalid documents and unexpected runtime exceptions.

The frontend interface was continuously improved to provide a better user experience through responsive design, progress indicators, loading animations, and enhanced report visualization.

Backend services were optimized to reduce execution time, minimize AWS resource consumption, and improve overall application responsiveness.

This iterative development approach ensured that the final system became more reliable, scalable, secure, and capable of delivering accurate AI-powered resume analysis while maintaining excellent performance under varying workloads.

3. TECHNOLOGY STACK

The **AI-Powered Resume Analyzer** was developed using modern web technologies, serverless cloud computing, and Artificial Intelligence. The selected technology stack ensures scalability, security, high availability, and efficient resume analysis while minimizing infrastructure management.

3.1 Frontend Technologies

React.js

React.js is used to build a responsive and interactive user interface. It enables users to register, log in, upload resumes, and view AI-generated analysis reports.

HTML5 & CSS3

HTML5 provides the structure of the web pages, while CSS3 is used to create a responsive and user-friendly interface.

JavaScript

JavaScript handles client-side functionality, API communication, and dynamic content updates.

3.2 Backend Technology

Python 3.12

Python is used to develop all AWS Lambda functions due to its simplicity, extensive libraries, and seamless integration with AWS services.

AWS Lambda

AWS Lambda executes the backend logic in a serverless environment. It handles resume upload, text extraction, parsing, AI analysis, report generation, and history retrieval without requiring dedicated servers.

3.3 Artificial Intelligence

Amazon Bedrock

Amazon Bedrock provides access to foundation models for intelligent resume evaluation. It analyzes resumes against job descriptions and generates ATS scores, job match percentages, strengths, weaknesses, and improvement suggestions.

Anthropic Claude

Claude is the AI model used through Amazon Bedrock to understand resume content and provide context-aware recommendations instead of simple keyword matching.

3.4 AWS Services

- **Amazon Cognito** – Manages user authentication and authorization.
- **Amazon S3** – Securely stores uploaded resume files.
- **Amazon Textract** – Extracts text from PDF and DOCX resumes.
- **Amazon API Gateway** – Provides secure REST APIs between frontend and backend.
- **Amazon DynamoDB** – Stores user details, resume data, and analysis reports.
- **Amazon CloudWatch** – Monitors Lambda functions and application logs.
- **AWS IAM** – Controls secure access and permissions between AWS services.

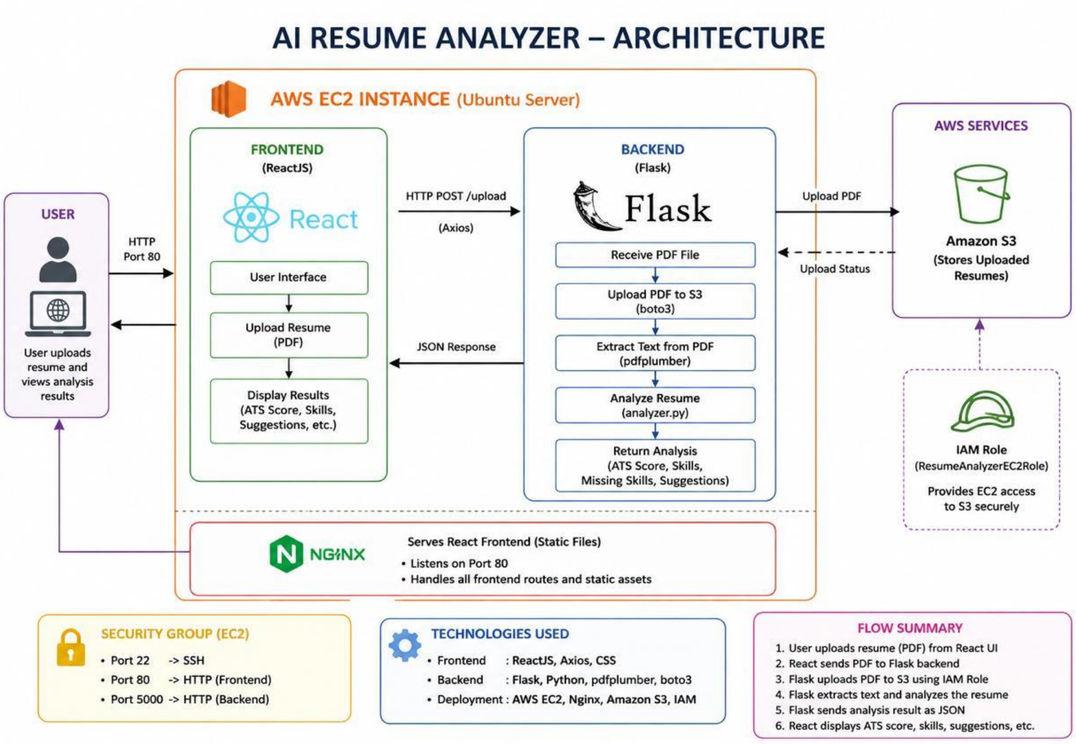
3.5 Development Tools

- **Visual Studio Code** – Source code editor.
- **AWS CLI** – Used to manage AWS resources.
- **AWS SAM CLI** – Used for building and deploying the serverless application.
- **Git & GitHub** – Used for version control and source code management.

4. SYSTEM DESIGN / ARCHITECTURE

The **AI-Powered Resume Analyzer** is designed using a **serverless cloud architecture** on Amazon Web Services (AWS). The system integrates multiple AWS services to provide secure authentication, resume storage, text extraction, AI-based resume evaluation, report generation, and history management. The architecture follows a modular and event-driven design where each component performs a specific responsibility, improving scalability, maintainability, and reliability.

The application consists of four major layers: the **Presentation Layer**, **Application Layer**, **AI Processing Layer**, and **Data Storage Layer**. Users interact with the React web application, which communicates with backend AWS Lambda functions through Amazon API Gateway. The backend services process uploaded resumes, invoke Amazon Bedrock for AI analysis, store analysis reports in Amazon DynamoDB, and return the results to the user interface.



4.1 Overall System Architecture

The overall workflow of the application is shown below.

Workflow

1. User uploads a resume through the React application.
2. Lambda generates a pre-signed URL.
3. Resume is uploaded securely to **Amazon S3**.
4. **Amazon Textract** extracts text from the resume.
5. Resume Parser extracts skills, education, projects, certifications, and experience.
6. Parsed resume and Job Description are sent to **Amazon Bedrock (Claude)**.
7. AI evaluates the resume and generates ATS score, job match percentage, strengths, weaknesses, and suggestions.
8. Analysis report is stored in **Amazon DynamoDB**.
9. Results are displayed on the frontend.
10. Users can view previous reports through the History page

4.3 Resume Upload – Amazon S3

The uploaded resume is stored securely in **Amazon S3**.

Instead of uploading files directly through the backend, the application generates a **pre-signed URL** using AWS Lambda. This allows the browser to upload the resume directly to S3 without exposing AWS credentials.

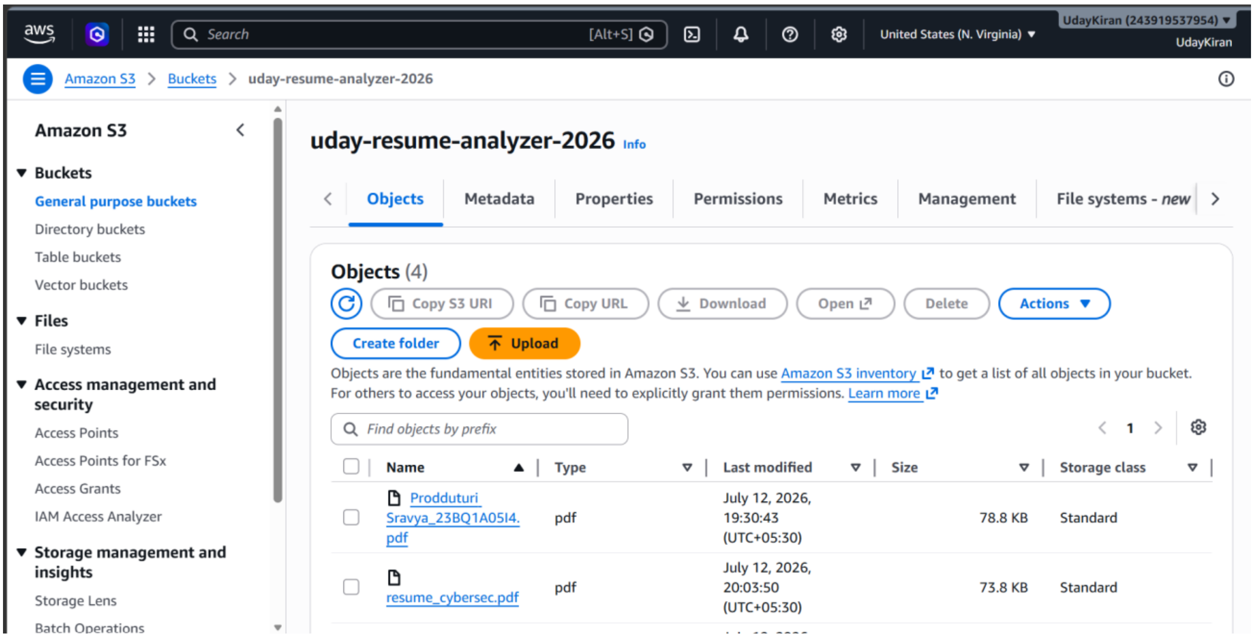
Supported file formats include:

- PDF
- DOCX

Using S3 ensures scalable, secure, and durable storage for uploaded documents.

Features

- Secure file upload
- Pre-signed URL
- High durability
- Automatic scalability



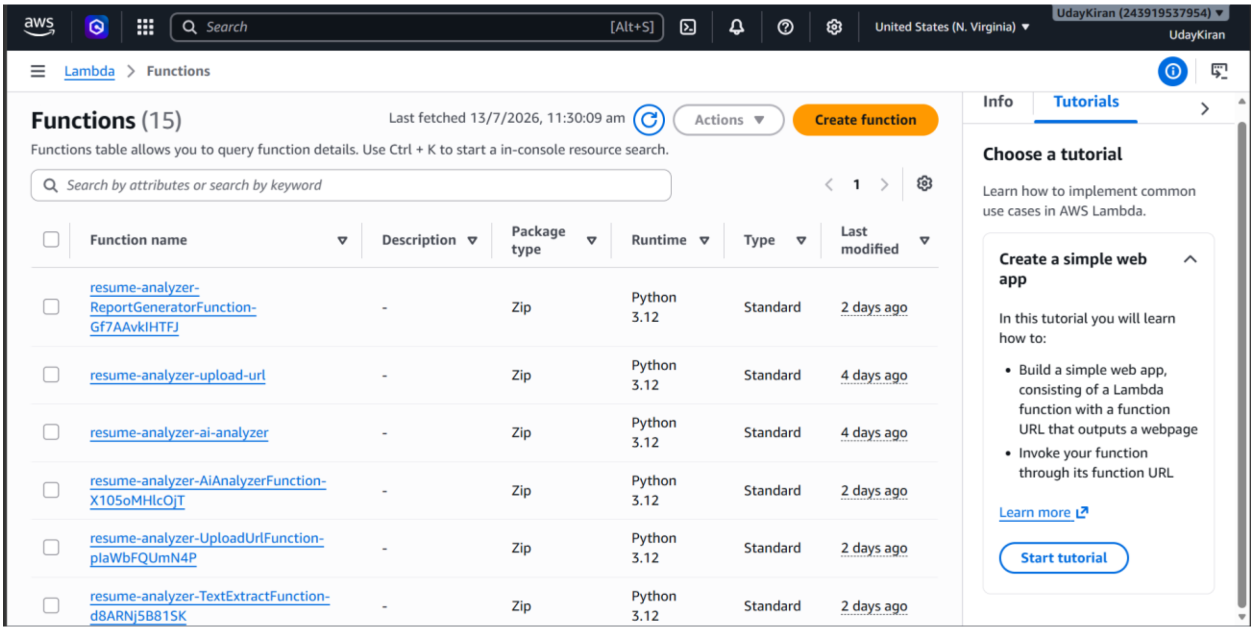
4.4 Resume Processing – AWS Lambda

The backend logic is implemented using AWS Lambda. Each Lambda function performs a dedicated task, making the application modular and easy to maintain.

The project contains the following Lambda functions:

- UploadUrlFunction
- TextExtractFunction
- ResumeParserFunction
- AiAnalyzerFunction
- ReportGeneratorFunction
- GetHistoryFunction
- CognitoPostConfirmFunction

This serverless architecture eliminates the need to manage servers while automatically scaling based on incoming requests.



4.5 Text Extraction – Amazon Textract

After the resume is uploaded to Amazon S3, the TextExtractFunction invokes **Amazon Textract** to extract textual information from the document.

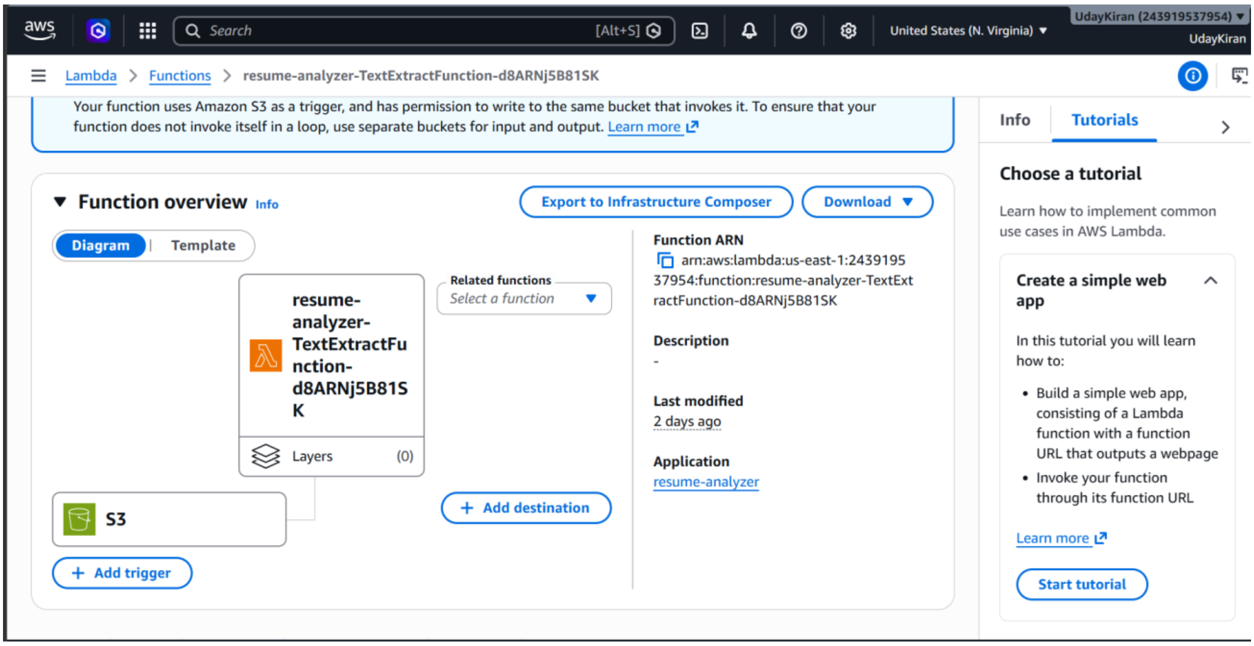
Textract accurately extracts:

- Personal Details
- Education
- Skills
- Experience
- Projects
- Certifications

This extracted text forms the input for the Resume Parser.

Features

- OCR Support
- Structured Text Extraction
- High Accuracy
- PDF and DOCX Support



4.6 AI Resume Analysis – Amazon Bedrock

The parsed resume and the job description are sent to Amazon Bedrock, where the Anthropic Claude foundation model performs intelligent resume evaluation.

Unlike traditional ATS systems that rely only on keyword matching, Claude understands the context of the resume and compares it with the job description.

The AI generates:

- ATS Score
- Job Match Percentage
- Matched Skills
- Missing Skills
- Resume Strengths
- Resume Weaknesses
- Improvement Suggestions
- Learning Recommendations

This enables users to improve their resumes based on AI-driven insights.

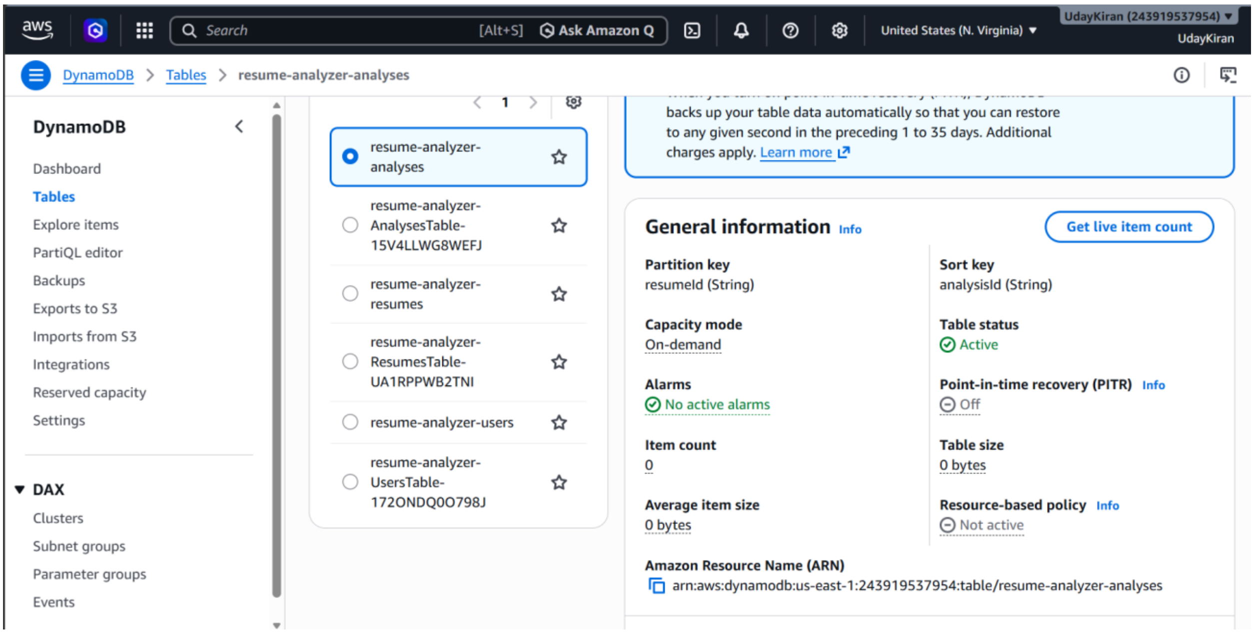
4.7 Database Management – Amazon DynamoDB

Amazon DynamoDB is used as the primary database for storing user information and resume analysis reports.

Each record includes:

- Resume ID
- User ID
- Upload Date
- ATS Score
- Job Match Percentage
- Skills
- Missing Skills
- AI Suggestions

DynamoDB offers high performance, automatic scaling, and low-latency data access.



4.8 REST API – Amazon API Gateway

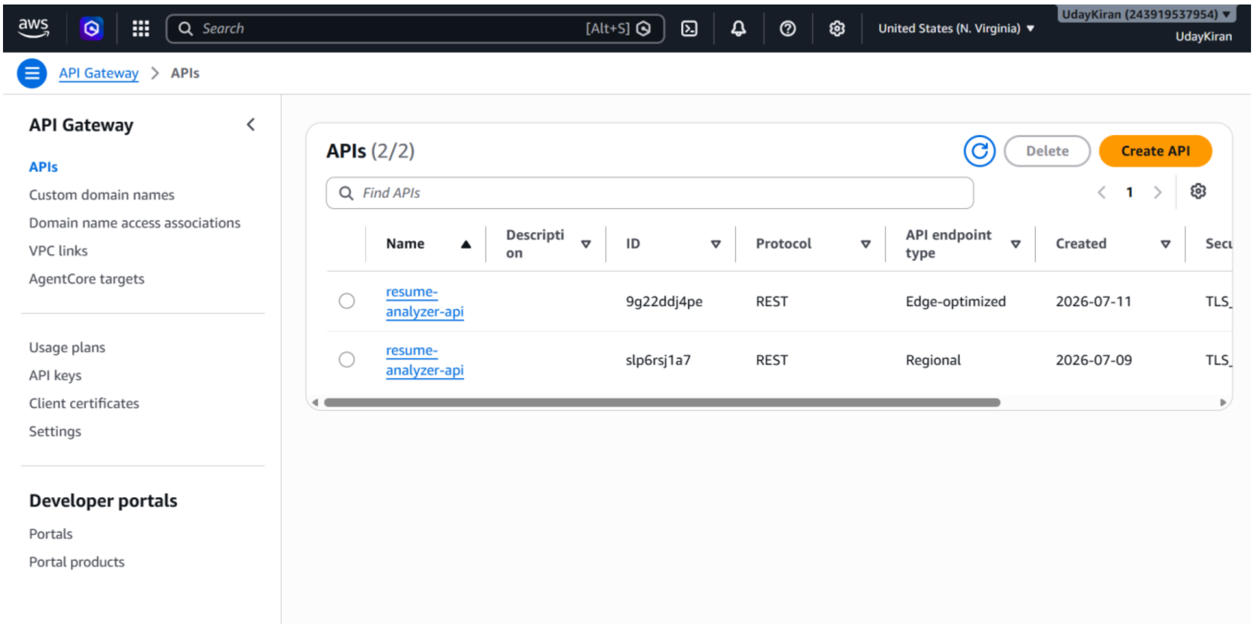
Amazon API Gateway exposes backend Lambda functions as secure REST APIs.

The frontend communicates with the backend through these APIs to perform operations such as authentication, resume upload, AI analysis, report generation, and history retrieval.

API Gateway validates JWT tokens issued by Amazon Cognito before forwarding requests to Lambda functions.

Features

- Secure REST APIs
- HTTPS Communication
- JWT Validation
- Request Routing



5. IMPLEMENTATION

The implementation of the **AI-Powered Resume Analyzer** focuses on automating resume evaluation using AWS serverless services and Artificial Intelligence. The application is developed using **React.js** for the frontend and **Python** for backend processing. AWS services such as Amazon S3, AWS Lambda, Amazon Textract, Amazon Bedrock, Amazon API Gateway, and Amazon DynamoDB work together to provide an efficient and scalable resume analysis system.

5.1 Resume Upload

The user uploads a resume in **PDF** or **DOCX** format through the web application. A pre-signed URL generated by AWS Lambda allows the resume to be securely uploaded to Amazon S3.

5.2 Text Extraction

Once the resume is uploaded, Amazon Textract extracts the textual content from the document. The extracted text is then prepared for further processing.

5.3 Resume Parsing

The extracted text is processed to identify important sections such as **personal details, education, technical skills, work experience, projects, and certifications**. This structured information is used for AI-based analysis.

5.4 AI Resume Analysis

The parsed resume and the provided job description are sent to **Amazon Bedrock**, where the Anthropic Claude model evaluates the resume. The AI generates an ATS score, job match percentage, matched skills, missing skills, strengths, weaknesses, and improvement suggestions.

5.5 Report Generation

The generated analysis results are formatted into a structured report and stored in **Amazon DynamoDB**. The report is then displayed on the web application, allowing users to view their resume evaluation in a clear and organized manner.

6. RESULTS

The AI-Powered Resume Analyzer was successfully designed, developed, and deployed using Amazon Web Services (AWS). The application performs automated resume analysis by extracting textual information from uploaded resumes, comparing it with a given job description, and generating an AI-powered evaluation report.

The system successfully provides an ATS Score, Job Match Percentage, Matched Skills, Missing Skills, Strengths, Weaknesses, and Improvement Suggestions. The serverless architecture ensures scalability, security, and efficient processing while minimizing infrastructure management.

The application demonstrates the effective integration of AWS services such as Amazon S3, AWS Lambda, Amazon Textract, Amazon Bedrock, Amazon API Gateway, and Amazon DynamoDB. The project successfully achieves its objective of providing an intelligent and cloud-based resume evaluation platform.

The developed application can be accessed through the following website:

Project Website:

http://100.55.8.138/

7. CONCLUSION

The **AI-Powered Resume Analyzer** was successfully designed and implemented using Amazon Web Services (AWS) and Artificial Intelligence to automate the resume evaluation process. The application provides an efficient platform for analyzing resumes by extracting text, comparing it with a job description, and generating a detailed analysis report.

The integration of AWS services such as **Amazon S3, AWS Lambda, Amazon Textract, Amazon Bedrock, Amazon API Gateway, and Amazon DynamoDB** enabled the development of a secure, scalable, and serverless application. The use of Amazon Bedrock improved the quality of resume evaluation by generating ATS scores, job match percentages, skill analysis, and personalized improvement suggestions.

The project successfully achieved its objectives by reducing manual effort in resume screening and providing users with valuable insights to improve their resumes. The serverless architecture ensures high availability, scalability, and cost-effective resource utilization.

In the future, the application can be enhanced by adding features such as recruiter dashboards, support for multiple languages, integration with job portals, interview question generation, and advanced career recommendations. These enhancements would further improve the usability and effectiveness of the system in modern recruitment processes.